

Puzzles to Puzzle You (Remastered)

Welcome to the tenth iteration of *Puzzles to Puzzle You (Remastered)* by Rani Rachavelpula, in which I will discuss the previous puzzle (Splitting Cherries) and pose the next one.

As an ode to a summer both beautiful and delirious, the cherry puzzle was meant to evoke sun-drenched picnics and long, wine-soaked dinners on the Mediterranean. But when the nostalgia fades and the spell of summer breaks, we discover that the puzzle is nothing more than the familiar combinatorial game of Nim dressed up in silk and sunscreen. Versions of the game have been played since antiquity—its lineage stretching back to the ancient Chinese pastime jiǎn-shízǐ, or “picking stones.”

Since its ancient debut, Nim has somewhat obtained a life of its own. At the 1939 New York World’s Fair, there was the Nimatron. Over a six month period, only a handful of fairgoers were able to defeat the electro-mechanical machine. Nim even appears in the French New Wave classic “L’Année dernière à Marienbad,” where one man repeatedly defeats another in this deceptively simple game, each victory an act of psychological one-upmanship in the pursuit of the enigmatic Delphine Seyrig.



*The game of nim in Alain Resnais 1961 film
“L’Année dernière à Marienbad”
(Last Year At Marienbad)*

As the Sprague–Grundy theorem reminds us, Nim lurks behind every impartial game. To know its inner logic is to hold a strange kind of power—the power to win, and occasionally, to beguile.

Puzzle #9 Splitting Cherries

In front of you and your friend are 4 bowls of cherries, containing, respectively, 5, 6, 7 and 8 cherries. You and your friend will alternately pick a bowl and take one or more cherries from it.

1. If you go first, and you want to be sure to get the last cherry, how many cherries should you take—and from which bowl?

Remove 4 cherries from the bowl of 8

2. If you go first, and you want to be sure your friend gets the last cherry, how many cherries should you take—and from which bowl?

Any other move
(eg. remove 1 cherry from bowl of 5).

Shoutout (Head of Strategic Investments at Citi) It is astounding to me how much this once-obscure, occult puzzling community has grown. Somehow, our latest challenge reached one of the most influential members at Citibank, who has requested anonymity. Not only did they solve it flawlessly, but they also reported an ‘almost spiritual’ rush of delight and wonder. That’s exactly why *Puzzles to Puzzle You (Remastered)* exists — proof that a good puzzle can move both minds *and* markets.

Solution The key to solving this puzzle is computing what is called the Nim-sum $\oplus$.

DEF: The Nim-sum of two or more nonnegative integers is the number obtained by taking their bitwise exclusive OR (XOR). That is, for integers $a_0, a_1, \dots, a_n \geq 0$,

$$a_0 \oplus a_1 \oplus \dots \oplus a_n$$

is their Nim-sum, where each bit is added modulo 2 without carrying.

The key is to recognize when the nim-sum is 0. If the Nim-sum is 0 at my turn, then my opponent has a winning strategy and there is nothing I can do to stop them (unless they make a silly mistake). In order to win, then, I need to make sure of two things:

1. The Nim-sum is non-zero at the time of my turn
2. I move in such a way that makes the Nim-sum 0 for my opponent's turn

Crucially, if the Nim-sum is not initially 0, it follows that I can always move in such a way that makes it 0. That means I have a winning strategy just in case the Nim-sum is non-zero at the start of the game. Let us check our bowls of cherries.

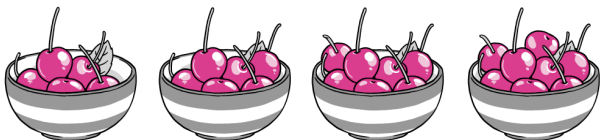
To calculate the Nim-sum associated with the cherry puzzle, first we convert all digits to binary. Since the Nim-sum $\oplus$ is associative (and commutative), we can calculate as follows:

$$\begin{aligned} 5 \oplus 6 \oplus 7 \oplus 8 &= (5 \oplus 6) \oplus (7 \oplus 8) \\ &= (0101 \oplus 0110) \oplus (0111 \oplus 1000) \\ &= 0011 \oplus 1111 \\ &= 1100 \\ &= 12 \\ &\neq 0 \end{aligned}$$

Since the Nim-sum is non-zero, we know there exists a winning strategy given that we are picking first. With some checking, one easily sees that the Nim-sum can be made to equal 0 by reducing the bowl of 8 cherries to a bowl of 4 cherries. One can also check by exhaustion that this is the only possible move, so any other move will give your opponent a winning strategy.

Thus, the solution to the puzzle is:

1. Remove 4 cherries from the bowl of 8
2. Any other move (eg. remove 1 cherry from bowl of 5)



Puzzle #10 Covering Stains

Just as a big event is about to begin, the queen's caterer notices, to his horror, that there are 10 tiny gravy stains on the tablecloth. All he has the time to do is to cover the stains with non-overlapping plates. He has plenty of plates, each a unit disk.

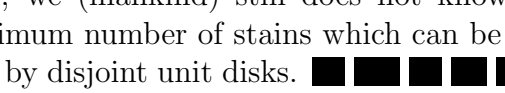
1. Can he do it, no matter how the stains are distributed?
2. How would you try to cover the stains if you had left your glasses at home, and couldn't see where the stains were?



Queen Elizabeth II, accompanied by Sir Hugh Roberts, the Director of the Royal Collection

This puzzle comes to us from my dear friend Wei. A budding mathematician and philosopher, Wei encountered this problem on a problem set in an advanced statistics course. I admit, when I first heard it, I thought Wei was mocking the level of this newsletter. It seemed to me so obvious that the answer to such a question is yes.

Then Wei asked me to prove it. This stumped me. As it turns out, one needs some nontrivial statistical methods to prove whether or not one can cover 10 stains for *all possible locations* of the stains.

This is no easy task, and I am currently brushing up on my statistics to tackle it. Indeed, we (mankind) still does not know the maximum number of stains which can be covered by disjoint unit disks.  How many stains are too many? Hopefully, one day we will know for sure.